

Why Only in the Lab? Understanding Counselors' Technology Acceptance Intentions on Virtual Reality Exposure Therapy: A Mixed Methods Study

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Abstract

The purpose of this study is to explore the factors influencing psychological counselors' acceptance of Virtual Reality Exposure Therapy (VRET), an emerging tool in clinical practice. Given the growing interest in VRET as an effective treatment for anxiety disorders, understanding the barriers and facilitators of its adoption is essential. We employed a mixed-methods approach that integrated a UTAUT-based quantitative analysis with a qualitative study using thematic analysis. The results revealed that performance expectancy ($\beta=.411$) and social influence ($\beta=.372$) were the primary factors driving counselors' intention to adopt VRET ($P<.001$). Moderators such as job stress and resistance to change also influenced these relationships. Qualitative responses highlighted the necessity of robust training and institutional support for successful implementation. Findings suggest that fostering belief in VRET's therapeutic effectiveness, reducing burdens through structured resources, and supporting professionals under high job demands may enhance clinical adoption of VRET.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**.

Keywords

Virtual Reality Exposure Therapy, Counselor Acceptance, Technology Adoption in Healthcare

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1 Introduction

Exposure therapy is widely acknowledged to be one of the most effective treatments for anxiety disorders [40]. Traditional exposure therapy repeatedly exposes patients to feared stimuli to desensitize their anxiety responses [33]. However, conventional exposure therapy often has practical challenges owing to time and financial constraints, making it difficult to reproduce real-life scenarios [29]. Moreover, controlling stimuli in a therapeutic setting can depend heavily on the therapist's competencies, which may pose challenges to the creation of precise and standardized treatment environments [15].

Virtual reality (VR) technology, delivered through head-mounted displays, offers immersive experiences that closely resemble real-world scenarios. Virtual reality exposure therapy (VRET) uses VR to create a safe yet engaging environment that induces anxiety-related stimuli, building on traditional exposure therapy principles to desensitize fear responses [43]. For instance, treating acrophobia no longer requires physical ascending to heights; VR can provide the appropriate stimuli. Therefore, VRET has garnered attention for its demonstrated safety and efficiency [25, 43, 44]. Research has shown that VRET can be as effective as traditional in vivo exposure therapy for anxiety disorders, including phobias and PTSD [31]. Additionally, the high level of control over exposure parameters in VR allows therapists to gradually increase difficulty in a structured manner, enhancing treatment efficacy [7]. As VR technology advances, its role in mental health interventions continues to grow, offering new possibilities for scalable and accessible treatment [1].

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Nevertheless, several barriers remain in implementing VRET in clinical practice. Previous research suggests that limited trust in new therapeutic technologies and a lack of competency in applying them are key obstacles, although such conclusions are often drawn from small-scale samples that restrict their broader applicability [27]. Furthermore, it did not address the specific barriers experienced by counselors in different clinical settings [27]. To successfully adopt new therapeutic technologies into clinical practice, it is important to explore the various factors that influence technology acceptance beyond therapeutic effectiveness [2, 41]. Consequently, it is crucial to identify the practical challenges experienced by counselors in diverse clinical environments and examine the factors influencing technology adoption [22]. Although previous research has extensively examined the clinical effectiveness of VRET, there remains a significant gap in understanding the factors influencing therapists' willingness to adopt the technology in real-world practice. While technology adoption models such as the Unified Theory of Acceptance and Use of Technology (UTAUT) have been applied in healthcare [3], their application to VRET adoption among therapists remains underexplored. Thus, this study focused on counselors working in settings such as not only university hospitals, but also public institutions, corporate counseling centers, and private clinics, and conducted an in-depth exploration of how they perceive VRET and the factors that shape their willingness to adopt it.

This study investigated the key factors affecting the acceptance of VRET among psychological counseling professionals by integrating quantitative model testing with thematic analysis. First, a quantitative model was derived from the UTAUT. This theoretical framework posits that performance expectancy, effort expectancy, social influence, and facilitating conditions collectively influence user intentions to adopt a technology [41]. Notably, UTAUT is also considered valid for understanding the technology acceptance behaviors of healthcare professionals, including telemedicine technologies [35]. And it has the flexibility to add or modify factors based on the characteristics of a particular demographic or technology. Building on prior findings regarding the conservative nature of healthcare contexts [37] and the detrimental effects of work-related stress on counseling outcomes [10], this study explores how these factors might shape the relationships between the UTAUT constructs and counselors' behavioral intentions. Specifically, we focus on the roles of job autonomy and job demands—key sub-factors of job stress—as moderating variables, operating under the assumption that contextual factors, such as these, could influence the relationship between performance expectations and behavioral intention in the adoption of VRET. Second, open-ended data were collected to analyze the participants' perspectives on VRET and gather contextual examples. Using a thematic analysis, this study examined counselors' attitudes and perceptions of VRET to identify the essential elements required for the effective implementation and institutionalization of VRET in counseling.

2 Related Works

2.1 The Need for Therapist-Centered Research in VRET Implementation

Existing studies have primarily focused on evaluating the clinical efficacy of VRET and its applicability from a patient-centered

perspective [18, 32]. However, these studies have not sufficiently addressed the perspectives of therapists, who are the key agents in the actual adoption of this technology in clinical settings.

In the field of technology acceptance research, the UTAUT model has been widely employed to examine the adoption of health technologies [3, 35]. This model identifies key determinants influencing adoption, including performance expectancy, social influence, and facilitating conditions [41]. Ong et al. (2024) conducted a survey-based study examining therapists' perspectives on telehealth-based VR therapy, revealing that security, cost burden, insurance coverage, and patient accessibility are critical considerations for adoption [26]. While these factors align with key constructs of the UTAUT model to some extent, the study primarily relied on quantitative survey data, which limits its ability to provide an in-depth exploration of therapists' actual adoption processes and perceived barriers.

2.2 Challenges of contextualizing technology adoption in healthcare

Decisions on whether to adopt new technologies depend not only on functional advantages but also on how well these technologies align with a counselor's professional values [23]. Moreover, as technological advancements reshape counselling and psychotherapy, practitioners respond diversely, with some embracing the opportunities to enhance practice and others expressing skepticism and apprehension about disruptions to traditional methods [4]. These nuanced perceptions are rooted in complex cognitive and contextual processes that often require in-depth qualitative methods. While qualitative research excels at exposing the lived experiences and opinions of healthcare providers [36], it can be limited by issues of generalizability and the potential for researcher bias [11, 16]. Accordingly, an integrated approach merging quantitative and qualitative methodologies offers a holistic perspective on technology adoption. Quantitative analyses highlight overall trends and structural relationships, whereas qualitative work provides deeper insights and context-specific details. Together, these approaches accurately capture the multifaceted nature of technology adoption in clinical settings [5].

3 Method

3.1 Overall Study Design

3.1.1 Participants. This study recruited certified psychotherapists aged 20–65 years old. Only practitioners with national licenses were permitted to participate, thus securing a minimum level of expertise. Participant recruitment was conducted through invitation emails sent to addresses listed in the open-access forums of the Korean Clinical Psychology Association and Korean Counseling Psychological Association, supplemented by posts shared on social media platforms by research team. Data were collected via a Google Forms survey, which included a brief overview of study objectives, confidentiality guarantees, informed consent, and a 4-minute introductory video explaining VRET. Each participant received a 10,000 KRW (approximately US\$7) gift card as compensation. Ethical clearance for the study was granted by the Institutional Review Board (IRB) of Ulsan National Institute of Science and Technology under Protocol UNISTIRB-23-057-A, confirming adherence to ethical guidelines and safeguarding the rights and welfare of participants.

The optimal sample size was calculated using the G*Power software with a medium effect size (Cohen's $f^2=0.15$, $\alpha=.05$, and $\text{power}=.80$). Considering the 12 predictive variables, including 4 key UTAUT predictors, age, gender, two job stress factors, and 4 Resistance to change factors, the recommended minimum sample size was 127. We aimed for a total of 254 participants, considering subgroup comparisons by gender and age.

3.1.2 Procedure. Data were collected remotely using a cross-sectional design. The survey first gathered demographic information, including gender, age, counseling experience, professional licenses, and typical client age groups. To account for the possibility that some counselors might not have been familiar with VRET, a 4-minute instructional video prepared by the research team was provided. The video explained five major elements: 1) the definition of VRET, 2) its intended applications, 3) sample content, 4) usage demonstrations, and 5) clinical evidence, thereby ensuring that the participants had sufficient context for VRET.

In the clinical evidence section, the following research findings were introduced. Studies have shown that Clinicians' attitudes toward VRET became more positive after experiencing it [34]. Additionally, VRET can provide a safe training environment, allowing users to practice without being exposed to actual risks [13]. Furthermore, its effectiveness increases when used alongside traditional treatments [20]. Lastly, research suggests that VRET is a more practical alternative compared to real-world exposure [6]. A snapshot of the video is shown in Figure 1. After viewing the video, the participants completed a survey, which culminated with open-ended items to solicit their opinions on VRET's feasibility and potential areas of improvement.

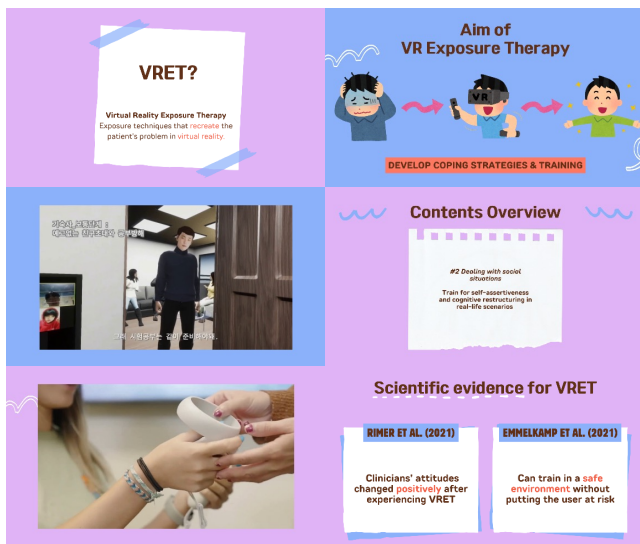


Figure 1: Overview of VRET introduction video

3.2 Study 1: UTAUT-based Technology Acceptance Model

3.2.1 Measures. Items relevant to the UTAUT model were adapted to the study context. The primary outcome variable, behavioral

intention, was measured using three items which evaluated the intention to integrate VRET into clinical practice. Performance expectancy assessed expectations about how VRET might improve therapy outcomes, while effort expectancy measured the anticipated level of effort for both the therapist and patient. Facilitating conditions captured the available resources for using VRET (cost, technical support, and infrastructure), and social influence evaluated the perceptions and opinions of patients, colleagues, and professional associations concerning VRET.

Resistance to change and job stress were moderators. Resistance to change has been identified as frequently reported psychological barriers to the adoption of new technologies in healthcare and counseling [17]. For counselors, resistance can be caused by the resistance to VR technology, which is at odds with traditional face-to-face techniques, or by the pressure to change the existing flow of counseling. Therefore, this study extends the concept of voluntariness in the UTAUT model to include resistance as a moderating variable. Job stress is known to cause counselors' professional and emotional burnout [42], which affects their ability to learn new skills and interact with patients [12]. Adopting a technology such as VRET requires additional training, equipment operation, and management, which may decrease acceptance intentions in an already high-stress environment. Therefore, we included job stress (specifically job demand and job control) as a moderator variable in this study.

The Resistance to change Scale assessed individual resistance to new practices, factoring in routine seeking, emotional responses, and short-term thinking [28]. Job stress was measured using subscales extracted from the Korean Occupational Stress Scale (KOSS), focusing on high job demand and low job control [39].

3.2.2 Data analysis. A stepwise multiple regression analysis was performed as the primary statistical method. After evaluating the correlations between the independent and dependent variables, a forward selection approach was adopted, incorporating factors based on their significance. Subsequent analyses tested the moderating effects of gender, age, job stress, and resistance to change. All regression analyses were conducted on Google Colab using the scikit-learn library. The resulting research model is illustrated in Figure 2.

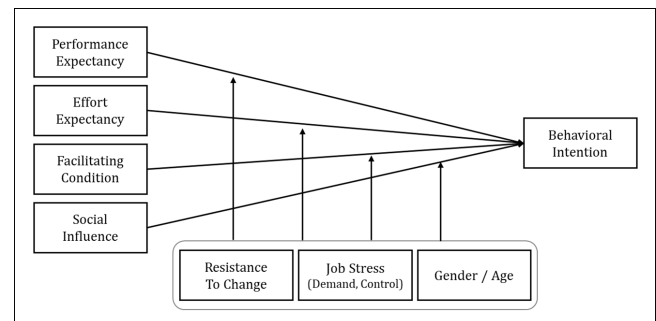


Figure 2: Multiple linear regression research model with moderator

3.3 Study 2: Thematic Analysis of Open-ended Responses

3.3.1 Open-ended questions. Open-ended questions were based on a 4-minute video. Participants provided free-text responses concerning their general views on VRET, along with perceived improvements and the feasibility of content targeting panic and social anxiety disorders. The video contained approximately one minute of sample content from the VRET materials, developed by the research team.

3.3.2 Data analysis. Text responses to open-ended questions were analyzed thematically using QDA Miner Lite (v. 3.0.6), a qualitative analysis tool developed by Provalis Research. Researchers initially conducted inductive coding by segmenting the participants' responses into meaningful units to build a preliminary thematic framework. This study adopted a consensus-based coding approach, where coding was conducted through iterative discussions until a consensus was reached [9]. During this process, the codes were refined and the relationships among the codes were defined. A structured codebook and coding guidelines were developed to ensure consistency within the research team.

In this study, we analyzed qualitative responses from 258 counseling professionals regarding the adoption of medical innovations, selecting only codes mentioned by at least 4% of participants as key themes. This approach aligns with Saldaña's (2012) emphasis on identifying recurring patterns in qualitative coding [38], ensuring the retention of meaningful themes while maintaining a clear analytical focus. Braun et al. (2022) analyzed barriers and facilitators of technology-assisted interventions using interview data from 41 participants, excluding codes reported by less than 10% of respondents [8]. However, our study involved a larger sample of 258 counseling professionals and specifically aimed to explore therapeutic innovations that have not yet been widely adopted. Given that technology adoption is influenced by a variety of factors, we set the threshold at 4% to ensure a more comprehensive analysis. This decision strikes a balance between excluding overly rare perspectives and capturing a diverse range of viewpoints, allowing us to identify key factors that professionals commonly recognize in the adoption process of therapeutic innovations.

4 Findings 1: Quantitative Insights

4.1 Demographics

Data from 258 counseling professionals were used for the thematic analysis. Of these, 86.8% (n=224) were female, with a mean age of 42.17 years (SD=8.84). More than half (63.2%, n=163) reported ten or fewer years of counseling experience, and the largest segment (39.2%, n=101) worked primarily with young adults (ages 19–29).

4.2 Feature selection

The stepwise regression analysis revealed that performance expectancy ($\beta = .404$) and social influence ($\beta = .387$) significantly affected behavioral intention ($p < .001$, 95%CI $-.452$ to $-.054$). In contrast, the change resistance variables and job-related factors (high job demand, low job control) were not statistically significant in explaining behavioral intention. Consequently, the final model consisted primarily of performance expectancy and social influence,

yielding $R^2 = .494$. Thus, these two predictors emerged as the principal factors for understanding and forecasting behavioral intentions.

4.3 Multiple linear regression with moderation effect

Multiple linear regression analyses were used to examine potential moderation effects. Routine seeking, low job control, and high job demands emerged as significant moderators and were incorporated into the final model. The analysis indicated that routine seeking and low job control moderated the effect of performance expectancy, whereas high job demand moderated the effect of social influence on behavioral intention. The final model attained $R^2 = .534$, reflecting an improvement in explanatory power with the inclusion of the moderators. The mean squared error (MSE) was .489, and the model's overall fit was $F(11,246) = 25.62$ ($p < .001$). These findings highlight that the identified moderators do indeed exert influence on the primary predictors (performance expectancy and social influence) of behavioral intentions. In particular, routine seeking and low job control emerged as critical factors moderating the impact of performance expectancy, and high job demand adjusted the effect of social influence. The final model is illustrated in Figure 3.

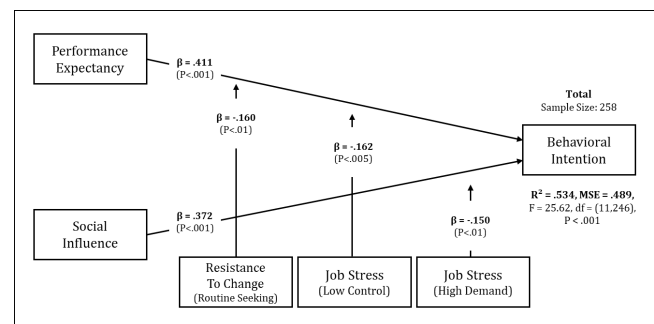


Figure 3: Main findings of the stepwise linear regression model with moderators

5 Findings 2: Qualitative Insights

5.1 Ambivalence toward VRET: Treatment or tool?

5.1.1 Scientific evidence on the effectiveness and safety of VR therapy is needed. Counselors indicated uncertainty regarding VRET due to insufficient empirical data establishing its efficacy. They stressed the need for evidence-based validation before clinical application. One participant stated, “Distinct evidence of effectiveness compared to other treatments is needed (P176),” emphasizing the necessity for clear comparative data demonstrating VR’s advantages. Another noted, “VR content seems promising, but more rigorous verification might be needed for social anxiety treatment (p195),” highlighting the importance of validating specific content. Concerns about how effectively and safely VR content can induce panic in clients have also surfaced, highlighting the need for robust evidence regarding safety and efficacy. There is an overall consensus that a well-substantiated rationale for VR beyond traditional methods must be clearly communicated.

5.1.2 Training or interaction that can be implemented alongside VR exposure. Participants noted that VRET should supplement, rather than replace, traditional counseling techniques. One counselor stated, “*It would be better if VRET included techniques like assertion training or mindfulness for anxiety relief* (P46),” underscoring the benefit of integrating psychotherapeutic skills with VRET for optimal anxiety management. Another participant advised, “*Rather than only providing problematic situations, it would be great to have calming scenery such as mountains or oceans for relaxation* (P35),” highlighting the need for stability-oriented content. Such integrative approaches have the potential to mitigate stress and anxiety in patients.

5.1.3 VRET as practice for real exposure therapy. VRET was perceived as an effective tool for safely preparing patients before real-life exposure. One participant noted that “*VR content could be useful as a simulation practice for clients before they transition to experiences with counselors or others* (P57),” describing VRET as a stepping stone in in-vivo scenarios. Another commented that “*VR can introduce a milder trigger for social anxiety than real-world challenges, allowing for a gradual approach* (P69).” Similar opinions held that “*VR could be used in the imagination and virtual exposure phases, setting the stage for real exposure* (P9),” underscoring VR’s practical value in gradual exposure frameworks.

5.2 Barriers and Facilitators for Introducing VR in Counseling Practice

5.2.1 Current resistance to unfamiliar technologies like VR. VR remains an unfamiliar technology for many counselors, potentially creating resistance. One participant observed, “*The concept of a virtual world evokes a lot of resistance. Reducing this resistance should be prioritized before implementing VR therapy* (P116).” Another opined, “*It will be crucial to improve awareness among both clients and stakeholders* (P73),” emphasizing the need to foster positive perceptions of VR. Specifically, one counselor remarked that “*Younger clients may be more open-minded, but both counselors and clients in middle or older age groups may resist* (P133),” highlighting generational disparities in technology acceptance. To overcome this problem, educational outreach and programs promoting initial adaptation are essential.

5.2.2 Essentiality of ongoing content development, distribution, and training. Sustained updates to both VR content and counselor education are critical for the long-term success of VRET. One counselor emphasized, “*It would be great if the content could be continually updated with various scenarios* (P132),” highlighting the importance of variety and ongoing innovation. Another said, “*Preliminary training, continuing education, and consistent management are necessary* (P85),” underscoring the role of training and oversight in maximizing the utility of technology. Calls for periodic upgrades to content and systems mirrored concerns about maintaining the efficacy and credibility of VRET over time. These efforts will lay the groundwork for successfully adopting VRET at a scale beyond pilot implementations.

5.3 VRET Requires Seamless Control for Counselors

5.3.1 Counselor training and pre-experience are required for effective delivery of VRET. Appropriate mastery of the technology by counselors was deemed crucial for effective VRET implementation. One participant noted, “*Mastery in applying VRET and professionalism in effectively using it with clients are essential* (P207),” highlighting the need for both technical proficiency and clinical integration skills. Another stated, “*A structured training program would be necessary* (P112),” reinforcing the call for comprehensive counselor education. Additionally, one participant observed, “*If there were workshops or gatherings where counselors could try out VR in advance, they would be more likely to adopt it* (P24),” suggesting that tangible first-hand experiences could increase willingness to use technology. Such programs should also prepare counselors to handle potential technical malfunctions, enabling them to confidently introduce VRET to their clients.

5.3.2 Clear guidelines, manuals, and instructions are essential for VR use in counseling. Structured guidelines and manuals are indispensable for ensuring the consistent quality of VR-based interventions. One counselor specified, “*Structured manuals or training should be provided to maintain quality control in VR therapy use* (P238),” underscoring the need for standardized procedures. Another participant highlighted that “*Both the counselor and patient need to fully understand the content and how it can be used effectively for specific issues* (P177),” calling for clarity in protocol and application. “*If a manual detailing how VR can be used in psychotherapy is provided, I would be very likely to utilize it* (P46),” mentioned another, suggesting that clear guidelines can significantly drive adoption. Such manuals facilitate uniform standards and offer actionable strategies for handling unexpected situations, thereby promoting confidence and proficiency among practitioners.

Table 1: Demographic Information of Participants Whose Quotes Were Cited

Participants	Gender	Age	Experience of Counseling
P9	Female	41	10–15 years
P24	Female	44	10–15 years
P35	Female	44	1–5 years
P46	Female	49	1–5 years
P57	Female	37	10–15 years
P69	Female	47	1–5 years
P73	Male	48	1–5 years
P85	Male	31	5–10 years
P112	Female	53	Over 20 years
P116	Female	55	15–20 years
P132	Female	33	1–5 years
P133	Female	36	5–10 years
P176	Female	45	5–10 years
P177	Female	43	10–15 years
P195	Female	53	10–15 years
P207	Female	36	5–10 years
P238	Female	28	1–5 years

6 Discussion

6.1 Enhancing Adoption of VRET: Building Belief in Efficacy and Reducing Implementation

The successful adoption of VRET among counselors depends not only on their belief in its clinical efficacy but also on the ability to integrate it seamlessly into their practice without adding excessive burdens. Our findings highlight that performance expectancy and social influence are key motivators for adoption.

However, environmental constraints—such as high job demands, low job control, and resistance to change—moderate these effects, making it critical to address both psychological and practical barriers simultaneously. For counselors to trust and adopt VRET, compelling empirical evidence is necessary to demonstrate its efficacy compared to traditional therapies. Many practitioners remain skeptical about new technologies in clinical settings, and without robust data, hesitation persists.

Additionally, in workplaces where VR is still perceived as novel, endorsements from colleagues, institutions, and professional associations can play a crucial role in fostering trust. As more counselors witness successful treatment outcomes among their peers, confidence in VRET can grow, leading to a self-reinforcing cycle of broader acceptance.

However, belief in VRET's effectiveness alone is insufficient if counselors feel overwhelmed by implementation challenges. High workload and limited autonomy may prevent them from engaging with new technologies, even when they acknowledge potential benefits.

To address these barriers, structured guidance—such as dedicated training programs and detailed manuals—can help ease the transition. By minimizing the cognitive and logistical burden of adopting VRET, organizations can encourage acceptance even in high-pressure environments.

6.2 Why Did Counselors Consider Effort Expectancy and Facilitating Conditions Less Important?

Effort Expectancy and Facilitating Conditions did not significantly influence counselors' adoption of VRET. Generally, these two factors are considered key determinants in technology adoption [41]. However, in this study, it is likely that the novelty of VRET in the counseling field and counselors' unfamiliarity with the technology had a greater impact on adoption decisions than these factors.

According to previous research, as age increases, differences in technology perception become more pronounced, and individuals are more likely to assess whether the effort required to learn a new technology outweighs its perceived benefits [21]. Given the demographic characteristics of the participants, counselors may have prioritized the question, 'Is learning this technology truly worthwhile?' over the simple consideration Effort Expectancy.

Meanwhile, the reason counselors did not place significant importance on Facilitating Conditions may be related to the nature of their profession. Unlike physicians or nurses, who often work in multidisciplinary teams, counselors tend to work independently. As a result, even if external technical support is available, it may

not necessarily serve as a decisive factor in the adoption of new technology. Instead, counselors are more likely to consider how well the technology aligns with their counseling approach, rather than the availability of technical support, when deciding whether to adopt it.

Based on these findings, providing technical support alone may not be sufficient to facilitate the adoption of VRET. More effective approaches may include structured, hands-on training, sharing best practices among counselors, and expert-led educational programs. Rather than simply instructing counselors on how to use technology, creating opportunities for direct experience with VRET and addressing psychological barriers to adoption will be essential.

6.3 Hands-On Experience in VRET Adoption

Previous research indicates that prior technological experience plays a significant role in an individual's adoption of new technologies [24, 30]. In particular, for immersive technologies such as VR, past experiences can substantially influence users' attitudes and behavioral intentions toward the technology [34]. If participants in this study had previous exposure to VR, their prior experiences may have influenced the interpretation of the findings. For instance, those with positive prior VR experiences might have led to a higher acceptance of VRET, whereas those with negative experiences may have demonstrated greater resistance to change. Therefore, future research should assess participants' previous VR experience and examine potential differences in VRET acceptance based on these experiences.

In this context, the importance of structured training and prior exposure, as identified in the qualitative analysis, becomes even more pronounced. If participants had been given the opportunity to experience VRET firsthand, their expectations or apprehensions might have been reflected differently in the study results. Simply providing theoretical information may be insufficient for counselors to fully comprehend and actively adopt VRET.

Conversely, offering direct experiential opportunities could lead to different research outcomes. Thus, when considering the practical implementation of VRET, it is essential not only to provide theoretical knowledge but also to create opportunities for counselors to engage with and familiarize themselves with the technology. This approach would enable counselors to develop a clearer understanding of VRET's potential benefits, ultimately increasing the likelihood of its integration into clinical practice.

6.4 Limitation

This study has several limitations. First, participants' varying familiarity with VR may have influenced their responses. Although an introductory video on VRET was provided, engagement levels varied. Future research should ensure standardized VR exposure for all participants. Second, participants' prior experience with VR was not measured, which may have impacted their acceptance of VRET. Prior exposure to technology can shape individuals' attitudes toward new systems, influencing their willingness to adopt them [24, 30]. Future research should assess past technology use and VR familiarity to better understand these effects. Finally, reliance on open-ended survey responses limited the depth of qualitative

insights. Future studies should incorporate structured interviews to capture more nuanced perspectives on VRET adoption.

7 Conclusion

This study examines the key factors influencing the adoption of VRET among psychological counseling professionals through both quantitative and qualitative analyses. Performance expectancy and social influence were identified as significant determinants of adoption. While counselors recognized the therapeutic potential of VRET, they also highlighted major barriers, including the lack of empirical validation, the need for technical proficiency, and challenges in integrating VRET into existing counseling practices. Addressing these concerns requires more than merely providing access to the technology; it necessitates the development of structured training curricula, hands-on practice opportunities, and evidence-based guidelines to support its implementation.

The COVID-19 pandemic accelerated the adoption of teletherapy, demonstrating the potential of digital mental health solutions [45]. However, whereas teletherapy relies on widely accessible video conferencing tools, VRET remains in its early stages, requiring specialized equipment and additional training. Given that trust and rapport are fundamental to counseling [19], VRET must not be introduced as a mere technological innovation but rather integrated gradually and harmoniously into existing therapeutic frameworks. Overcoming technological barriers alone is insufficient; VRET must be designed and implemented in a manner that preserves the core principles of counseling without disrupting the therapeutic process. A blended approach, which integrates VR with traditional therapy, may help balance innovation with human connection [14].

Furthermore, previous research indicates that mental health care providers are more likely to use telehealth services frequently when provided with clear guidelines on how to successfully implement telehealth practices [45]. This underscores the importance of offering well-structured guidance and training to facilitate the effective adoption of VRET in clinical settings.

To establish VRET as a viable therapeutic tool, professional training must extend beyond theoretical instruction to encompass practical experience in real-world settings. This necessitates the development of formal educational curricula that equip counselors with the necessary skills to operate VR tools effectively, alongside continuous content updates informed by the latest research. Additionally, structured hands-on training programs should be introduced, enabling practitioners to gain direct experience with VR devices, alleviate technical apprehension, and build confidence in their application. Furthermore, mentorship initiatives should be established, fostering collaboration between experienced VRET users and novice counselors, while peer learning opportunities should be promoted to facilitate knowledge sharing and best practice development within the field.

For VRET to transition from an experimental tool to a widely accepted therapeutic modality, enhancing technological accessibility must be accompanied by comprehensive training programs, hands-on learning experiences, and strategies to maintain the integrity of the therapeutic alliance. With sustained institutional and policy support, VRET has the potential to become a scalable and effective

intervention within diverse clinical settings, contributing to the broader advancement of digital mental health care.

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A Questionnaire for Measuring the Elements of the UTAUT Model (5 point Likert Scale)

Construct	Survey Items
Behavioral Intention (BI)	If VR devices and exposure therapy content are provided, I would be willing to use them in psychological therapy. If possible, I would incorporate VR exposure therapy content into my therapeutic process. In the future, I intend to apply VR exposure therapy content with clients.
Performance Expectancy (PE)	VR exposure therapy content will be useful in the psychological therapy process. VR exposure therapy content will alleviate clients' anxiety-related symptoms. VR exposure therapy content will aid my work as a therapist. VR exposure therapy content will increase client satisfaction with psychological therapy. VR exposure therapy content will enhance clients' engagement in therapy.
Effort Expectancy (EE)	Most clients will have the necessary skills to use VR exposure therapy content. It won't be difficult for clients to learn how to use VR exposure therapy content. The information provided by VR exposure therapy content will be easy for clients to understand. Most therapy providers will have the necessary skills to offer VR exposure therapy content. It won't be difficult for therapy providers to learn how to use VR exposure therapy content. Charging and managing VR devices to provide VR exposure therapy content won't be difficult.* Providing various VR exposure therapy contents according to clients' symptoms and levels won't be difficult.*
Facilitating Conditions (FC)	I have the resources (budget, space) necessary to utilize VR exposure therapy content. If I encounter difficulties in using VR exposure therapy content, I will be able to seek help from others. VR exposure therapy content can be used alongside other therapeutic techniques I employ.*
Social Influence (SI)	Clients will not have a negative attitude toward the use of VR exposure therapy content. Psychological therapy-related organizations and associations will recommend the use of VR exposure therapy content. My peers in the therapy provision will recommend the use of VR exposure therapy content.